Carbon Footprint Assessment of Thai Cuisine: Curry and Rice Menus

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ABSTRACT

The demand for food production is expected to increase significantly as the global population continues to grow, placing pressure on the food system and the environment. The current food system is a major contributor to global warming and requires a transition towards more sustainable food production and consumption. This study assessed the carbon footprint of menu consumption, covering the emissions from ingredient acquisition to the cooking process. A total of 82 Thai curry and rice menus were analyzed and categorized by meat type, e.g., pork, poultry, fish, and cooking methods, e.g., frying, boiling, and steaming. The average CF was found to be highest in poultry-based dishes (3.34 ± 1.18 gCO_2e/g), followed by pork (3.21 ± 1.45 gCO_2e/g), and lowest in fish-based (1.47 ± 0.78 gCO_2e/g). In contrast, the average CF of non-meat dishes was 2.67 ± 1.97 gCO_2e/g. Key CF contributors varied by dish type, with meat and eggs being the most carbon-intensive ingredients. LPG consumption, which varied across different cooking methods, also influenced the CF emission. Modifying menu composition and being aware of LPG consumption in the cooking process could lessen the CF of the menu. The results reinforce the need for an integrated approach in the food industry that considers ingredients and cooking methods, aiming to develop sustainable food systems that align with global climate goals and enhance public health.

Keywords : Life Cycle Assessment; Carbon Footprint; Food production; Thai Cuisine; Sustainability

INTRODUCTION

Modern food systems encompass multiple stages, including production, transportation, processing, packaging, storage, retail, consumption, food loss, and waste management [1]. This system plays a critical factor for human health, and ensuring sufficient food supply is the foundation of well-being [2]. The United Nations projected that the global population would reach 9.7 billion by 2050, necessitating a 70% increase in food production to meet rising demand [3]. However, food production and consumption processes significantly contribute to environmental issues [4]. Current dietary should be improved to enhance sustainability and reduce environmental impacts. This includes minimizing food waste, which accounts for one-third of global food production. Ensuring food security while maintaining sustainability presents a significant challenge [5]. The Sustainable Development Goals, the second goal is advocating food security and the promotion of sustainable food production. [6]. Different dietary patterns have varying environmental impacts, emphasizing the need to shift consumption habits to mitigate these effects. Adopting more sustainable dietary choices can help reduce the environmental footprint of food production and contribute to long-term global sustainability [7].

The agricultural sector covers approximately 37% of the earth's land surface and is a major source of methane and nitrous oxide emissions, accounting for 52% and 84% of global anthropogenic emissions, respectively [8]. These are greenhouse gases (GHG) and lead to global warming crisis. Significant emissions for ruminants are explained mainly by methane emissions from enteric fermentation which makes animal-based food products significantly more impactful than plant-based alternatives in terms of their environmental footprint [9].

Given that the current food system is a major contributor to global warming, it is imperative to transition towards more sustainable food production methods and encourage environmentally responsible consumption behaviors. This study aims to evaluate the carbon footprint (CF) of meal consumption. The CF information can reveal opportunities for reducing the average carbon footprint of food products. Providing consumers with CF information on menus helps to encourage consumer decision-making in food selection to make climate-friendly food choice.

METHODOLOGY

This study was scoped into curry and rice menus. The menus were surveyed from food vendors at Chiang Mai University during August – September 2024. Then, the recipes and cooking time were extracted from an online cooking recipe-sharing platform called "Wongnai." All menus were categorized based on ingredients (e.g., with meat and without meat) and cooking methods (e.g., stirred-fried, boiled, etc.). The CF of menus was calculated covering the emissions from ingredient acquisition to cooking process (cradle to gate), using eq. 1. The calculation excluded the transport emissions of ingredients to kitchens.

$$CF = \frac{\sum_{i=1}^{n}(X_i \times EF_i) + Y_L \times EF_L}{X_T} \qquad \text{—eq 1}$$

Here, CF is the amount of carbon footprint of the menu (gCO_2e/g)., X_i is the weight of ingredient i used in the menu (g)., EFi is the emission factor of the ingredient i (gCO_2e/g)., Y_L is the weight of LPG used in the menu (g), EF_L is the emission factor of LPG including the emissions from acquisition and combustion in cooking process (gCO_2e/g). The weight of LPG used is calculated by multiplying the total cooking time in minutes by the LPG consumption rate. All cooking methods were assumed to have the same rate of 1.67 grams per minute, which was determined from the experiments conducted in this study. X_T is the total weight of ingredients (g). All emission factors were from the Thai National LCI Database, TIIS-MTEC-NSTDA and the TGO certified carbon footprint products. If the emission factors were unavailable in the national database, the study first considered whether the weight of the ingredient was less than 5% of the total weight. If so, the ingredient was excluded from the calculation. If the weight of the ingredient accounted for more than 5% of the total weight, a similar ingredient's emission factor was used instead. All calculations were done using Microsoft Excel version 2501 Build 16.0.18429.20132.

RESULTS AND DISCUSSIONS

1. Menu categories

A total of 82 menus were observed and categorized into meat-based and non-meat-based menus. The meat-based included 29 pork dishes, 21 poultry dishes and six fish dishes. The remaining 26 menus were non-meat based. Based on cooking method, the menus were classified into seven categories: 45 stir-fried dishes, 22 boiled dishes, nine fried dishes, two steamed dishes, two grilled dishes, one simmered dish and one baked dish.

2. CF of curry and rice menus

Regardless of the cooking method, poultry dishes exhibited the highest average CF (3.34 ± 1.18 gCO_2e/g), followed by pork (3.21 ± 1.45 gCO_2e/g), non-meat dishes (2.67 ± 1.97 gCO_2e/g), and fish (1.47 ± 0.78 gCO2e/g), as shown in Figure 1. The primary contribution to CF varies by meat-type. Chicken meat was the highest contribution in almost all poultry dishes (20 out of 21 dishes or 95%). Pork ranked the highest contribution, at 72% of the total pork-based dishes. For the fish-based dishes, LPG was the dominant contributor for 50% of fish dishes; the other 33% had their largest contribution from fish and 17% had from eggs. As for non-meat based, eggs showed the most significant emissions in 58% of the total non-meat-based dishes, instant noodles and glass noodles (15%), LPG (15%), and green vegetables (12%).

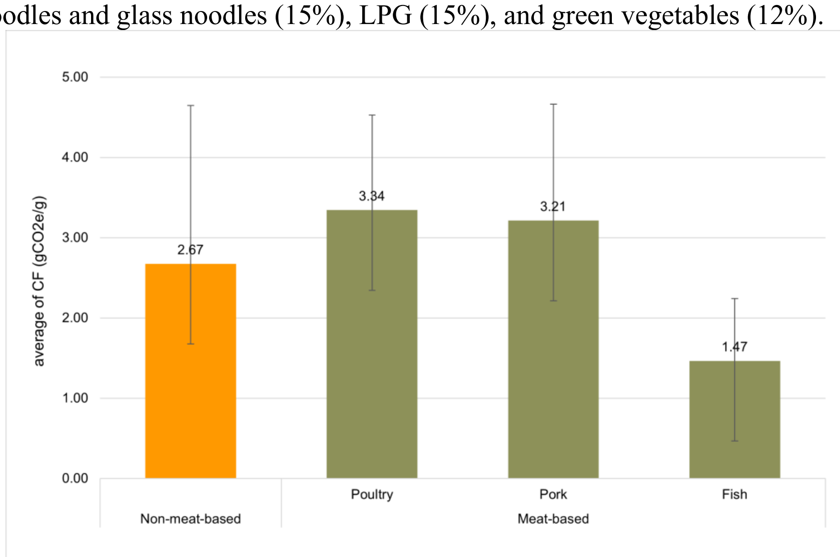

Figure 1 CF of meat-based and non-meat dishes

In meat-based dishes, the highest CF was found on the menu stewed pork and egg in brown sauce (6.49 gCO_2e/g), followed by and clear soup with egg and minced pork menu (6.09 gCO_2e/g). In contrast, Choo Chee curry with fish was the lowest CF menu (0.80 gCO_2e/g). The dishes with high CF typically contain pork and eggs, as shown in **Figure 2,** both are ingredients associated with high GHG. For non-meat based, the highest CF dish was sweet and sour fried boiled eggs (7.78 gCO_2e/g), while the lowest CF menu was stir-fried vegetables (0.49 gCO_2e/g). This indicates that eggs are a key contributor to GHG among non-meat dishes. Opting for a diet that choosing fish or vegetables as an alternative could contribute to lowering emissions.

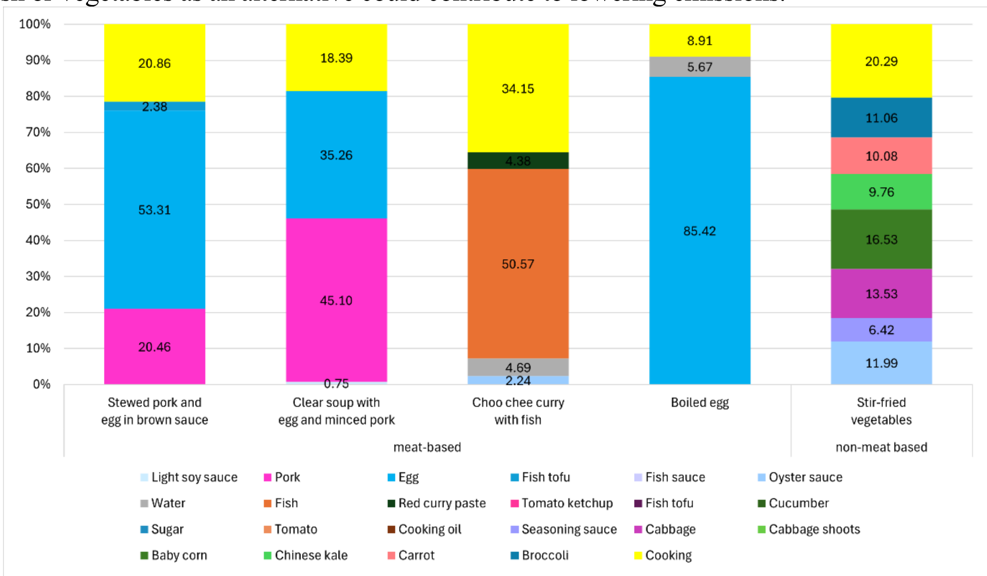

Figure 2 GHG contributing composition in some selected menus

Our findings are consistent with the previous studies. The study by Leuenberger et al. [10] reported that meat-based dishes generated higher GHG emissions, on average 3 $kgCO_2e$/serving, and poultry or pork ranged from 1.5 to 2 $kgCO_2e$/serving. While the vegetarian dishes had 0.9 $kgCO_2e$/serving. Another study by Volanti et al. [11] showed that the meat, fish and dairy dishes had high CF, ranging from 3 to 24 gCO_2e/g, with the highest value at beef. Fruit and vegetables had a smaller CF than 2 gCO_2e/g.

Considering the cooking methods, that consumed LPG varied, **Figure 3** presents that fried dishes had the highest CF (4.83 ± 1.83 gCO_2e/g), followed by steamed (4.28 ± 0.54 gCO_2e/g), grilled (3.66 ± 2.93 gCO_2e/g), stir-fried (2.52 ± 1.31 gCO_2e/g), boiled (2.92 ± 1.54 gCO_2e/g), simmered (2.22 gCO_2e/g), and baked (2.06 gCO_2e/g). Carefully considering the footprint contribution of LPG usage, it showed that grilled dishes had the highest contribution from LPG usage (55%), followed by baking (32%), boiled (24%), steamed (24%), stir-fried (20%), fried (17%), and stewed (10%). This result should be cautious, as there were few menus cooked steamed, grilled, simmered, and baked in our study, as described above.

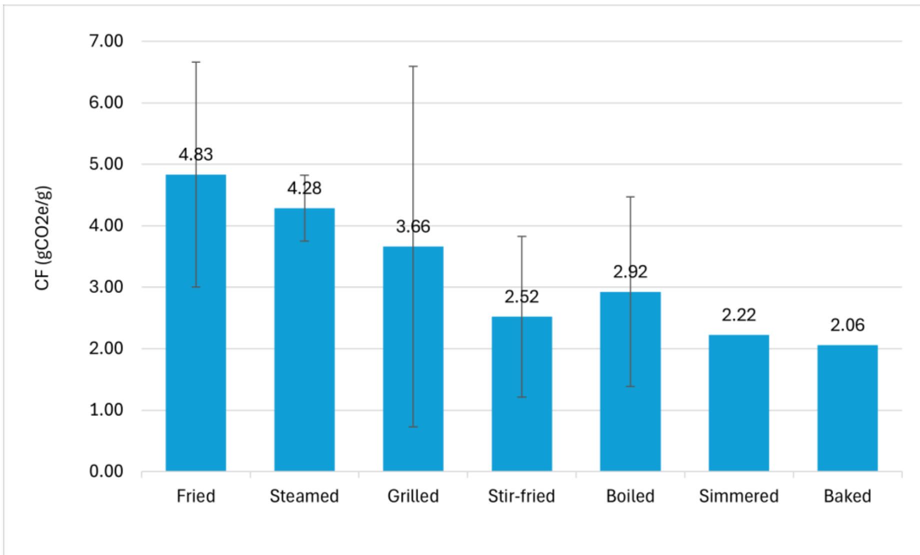

Figure 3 CF of menus by cooking methods

Our results show that meat-based dishes, particularly poultry and pork, had higher CF than fish and vegetarian options. This is consistent with global trends highlighting the environmental impact of meat production. By shifting towards plant-based diets, individuals can substantially reduce their carbon emissions. Introducing CF information on menus can influence consumer choices, encouraging shifts towards more sustainable options. Such transparency in menu labeling can drive significant behavioral changes, supporting broader climate change mitigation efforts. Additionally, our findings reveal that the LPG could be a significant CF contributor, such as fish dishes, where LPG accounts 50% of the total CF portion. The estimation assumed that all cooking methods use the same rate of LPG consumption, differing only in the cooking time.

CONCLUSION

The present study evaluated the GHG associated with different curry and rice menus. The emissions covered from raw material acquisition to cooking process. The findings indicated that meat-based dishes had higher CF. Shifting dietary behaviors can contribute to reducing greenhouse gas emissions. The outcome of the present study could raise consumers awareness of the need to encourage environmentally friendly menu choices, promoting sustainable food consumption.

The limitations of this study primarily stem from its restricted data collection scope and the inherent variability in recipes. Furthermore, the carbon footprint calculations did not encompass emissions associated with the transportation of ingredients to the kitchen, and food waste generation, e.g., from cooking and after consumption. Incorporating transportation and food-related emissions into the carbon footprint analysis would offer a more holistic view of the total environmental impact, guiding more targeted strategies for reducing GHG in the food sector. In addition, future investigations could expand the scope of data collection to include a broader array of recipes and culinary practices, enhancing the generalizability of the findings.

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